

Stephen P. Quartarone

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EDUCATION

Northeastern University, Boston, MA

Expected Graduation - 2028

B.S. in Mechanical Engineering, Minor in Aerospace Engineering | *GPA: 3.6*

Related Coursework: Fluids, Thermodynamics, Mechanics of Materials, Material Science, ME Computation & Design

SKILLS AND CERTIFICATIONS

Engineering Tools: CAD, C++, MATLAB, ANSYS, GD&T, Excel

Technical Skills: DFM/DFA, Fixture Design, Rapid Prototyping, Additive Manufacturing, Master Sketching

Certifications: Certified SolidWorks Associate (CSWA), Additive Manufacturing, Sustainability

WORK EXPERIENCE

Manufacturing Engineering Co-op, Globe Composite Solutions, Stoughton, MA

Jan 2026 - Jun 2026

- Delivered 18 engineering projects that generated \$657,438 in documented cost savings through labor, tooling, process, and manufacturing improvements using DFM
- Spearheaded manufacturing launch efforts for a new business line by designing molds, developing production processes, creating standardized work instructions, and communicated with vendors
- Designed, fabricated, and implemented manufacturing tooling using metal, wood, polyurethane, and commercial hardware, collaborating closely with machinists and operators on the production floor
- Led cross-functional project meetings and coordinated with engineering, manufacturing, purchasing, and leadership to drive projects from concept through implementation while navigating design and spec constraints

Construction Assistant - Single-Family Home, Wrentham, MA

July 2025 – Sep 2025

- Co-built a 900 sq ft single-family home alongside a retired construction professional; contributed to framing, exterior envelope, and finish work (40+ hrs/week)
- Applied SolidWorks to support design and measurements, modeled sunburst feature to visualize aesthetics

PROJECTS

Dual Cavity Mold Design, Globe Composite Solutions, Stoughton, MA

Feb 2026 - Jun 2026

- Led design of a dual-cavity mold that doubled production capacity & maintained existing floor space
- Facilitated design reviews with engineering and leadership, got approval for a \$600,000 production rollout
- Managed the project from concept through detailed CAD, supplier coordination and prototype planning
- Supported a manufacturing program projected to process over \$130 million in revenue over the next five years

3D Printed Push-Up Parallettes, Independent Project

Feb 2026 – Jun 2026

- Designed and constructed lightweight push-up parallettes, optimized for strength, stability, and printability.
- Performed FEA in ANSYS under a 228 N design load, identifying a maximum von Mises stress of 0.90 MPa

Northeastern Aerospace Club, Northeastern University, Member

Jan 2026 – Present

- Building a high-power rocket in preparation for Level 1 certification
- Learning rocket construction, flight operations, recovery systems, and launch safety procedures
- Attended weekly meetings to work on CAD for electronics bay in large scale rocket launch

Cornerstone Project - Warehouse Organizer Robot, Northeastern University, Boston, MA

Spring 2025

- Developed a warehouse robot prototype with a scissor lift mechanism by 3D-printing gears and lift
- Designed a robot to transport crates and improve warehouse efficiency with line tracking software

ACADEMIC COMPETITIONS AND AWARDS

Cornerstone Project - Sumo Robot, Northeastern University, Boston, MA

Fall 2024

- Earned 1st place in class-wide competition of 7 teams by programming tracking logic to push rivals out
- Built and programmed robot using 3D printing, Arduino, and C++ for optimized performance

Special Interests: Aerospace, Design, Boston-based films, McMaster-Carr, Renewable Energy